

BUSA8090 Data and Visualisation for Business

SESSION 1

ASSESSMENT TASK – 2



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Introduction

Making informed business decisions about marketing requires data.

Understanding current music trends may be beneficial for major music firms such as Sony Music, since it allows them to better serve their artists and expand their audience base. This paper focuses on analysing Spotify's most-streamed song statistics in 2023 using Tableau, an effective data visualization tool. We can cover the key trends and findings that can help Sony Music's Chief Marketing Officer (CMO) develop successful marketing plans by developing various visuals.

We have created a report and divided the work into two sections: The first section focuses on data visualization using Spotify. Every visual aid facilitates decision-making by highlighting significant patterns. We'll go over each visualization's creation process and the insights it offers into current trends in music streaming.

Additionally, there is comprehensive information about data analytics, data governance, ethical issues, and responsibilities in the second section.

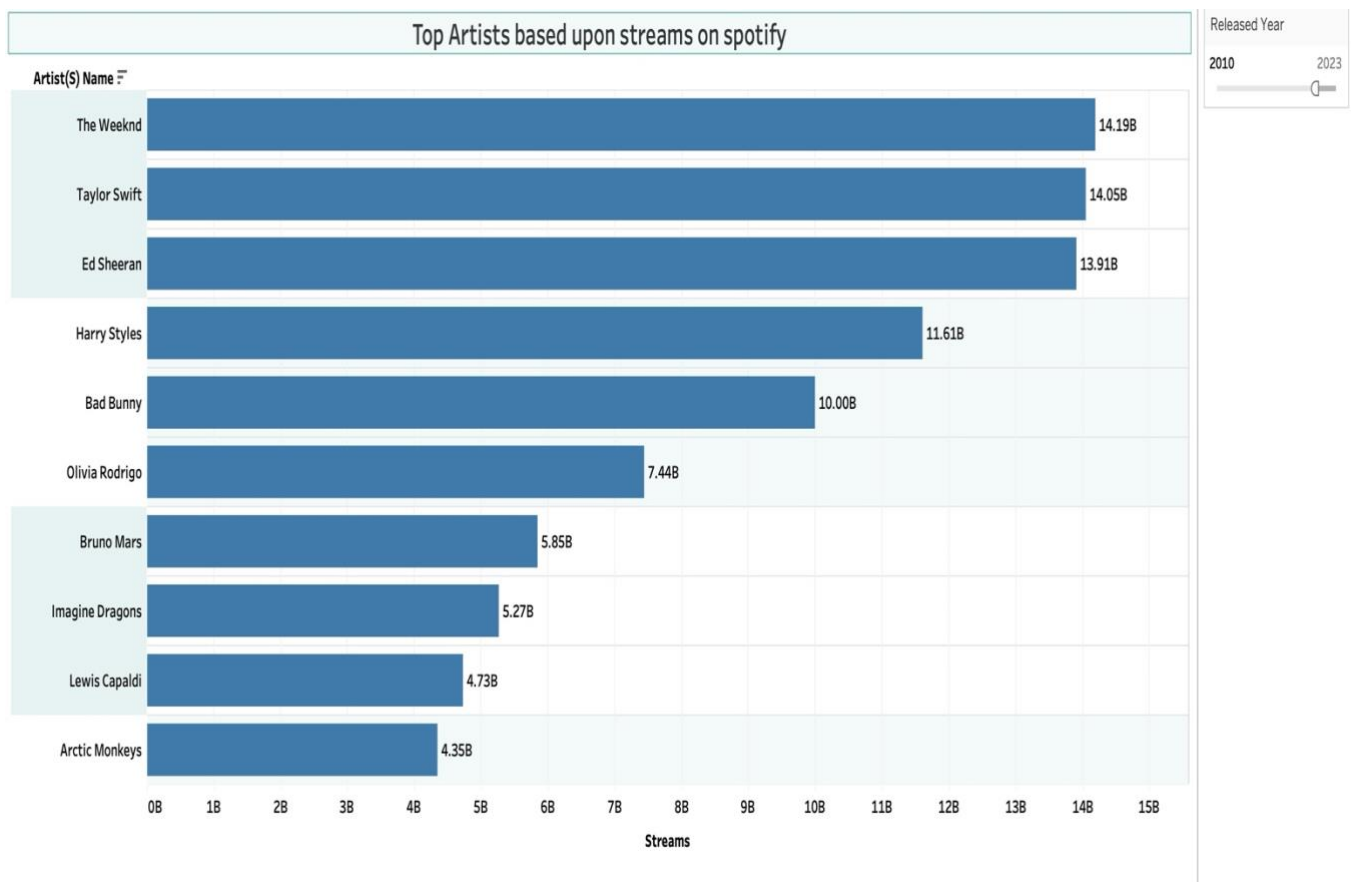
Furthermore, we provide explanations for the forecasts and recommendations derived from the visualizations. By the time I finish reading this study, I will understand how Sony Music can enhance their marketing tactics, forecast future trends, and make better business decisions with the use of data visualization.

Part A

VISUALISATION 1

Visualisation 1A: Top 10 artist based upon streams on Spotify.

Visualisation 1B: Top Artists combinations.



(1A from Tableau Workbook)

1. How I Did It:

- **Data Connection:** Connected to the dataset Cleaned_Spotify_Dataset.csv to generate the visualization.

- **Data Verification:** Ensured that data types for streams, released year, and artist(s)_name were set correctly.

Bar Chart creation:

- Rows-: Placed Artist(s) Name in the Rows shelf.
- Columns-: Placed SUM(Streams) in the Columns shelf to display the total number of streams for each artist.
- Computed Field and Parameter: Developed a computed field to rank artists based on their total streams.
- Created a parameter called Top N to filter and display the top artists.

Filtering:

- Used the computed field and Top N parameter on the Filters shelf to show only the top N artists.
- Added filters for Artist Count to enhance interactivity, allowing users to focus on specific years or artists with a particular number of tracks.

Chart Customization:

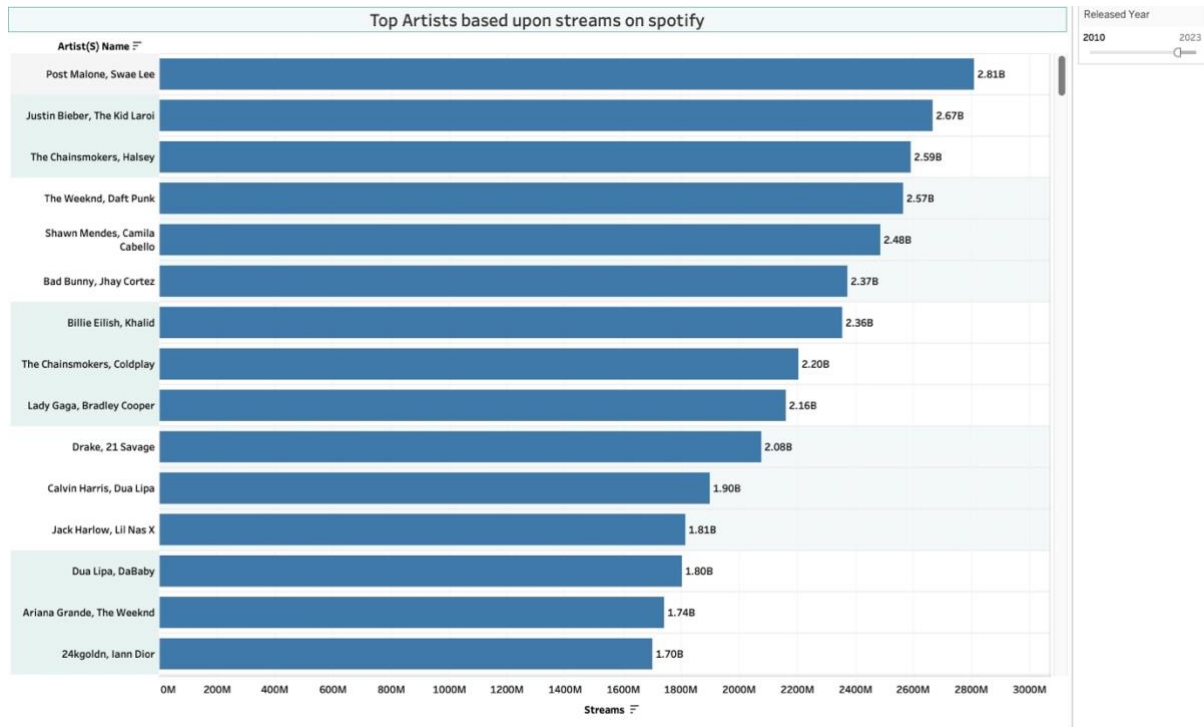
Titled the chart "Top Artists" to clearly indicate its content. Included a slider for Top N to dynamically adjust the number of top artists displayed. Applied a filter for Released Year ranging from 2010 to 2023.

(Ludovicocuoghi, 2021)

2. What It Tells Us:

- **Top Artist:** The most popular musicians on Spotify are highlighted in this visualization, which ranks them according to total streams. It demonstrates that some of the most widely streamed artists are The Weekend, Ed Sheeran, and Taylor Swift, with The Weekend having the most streams overall.
- **Streams Count:** More than 14 billion streams were gathered from weekend since 2010.

- **Top Artists Combination:** Another visualisation includes the filter with artist count as 2. When SONY wants to have 2 artists then on which combination SONY should put its money to get the best results.



(1B from Tableau Workbook)

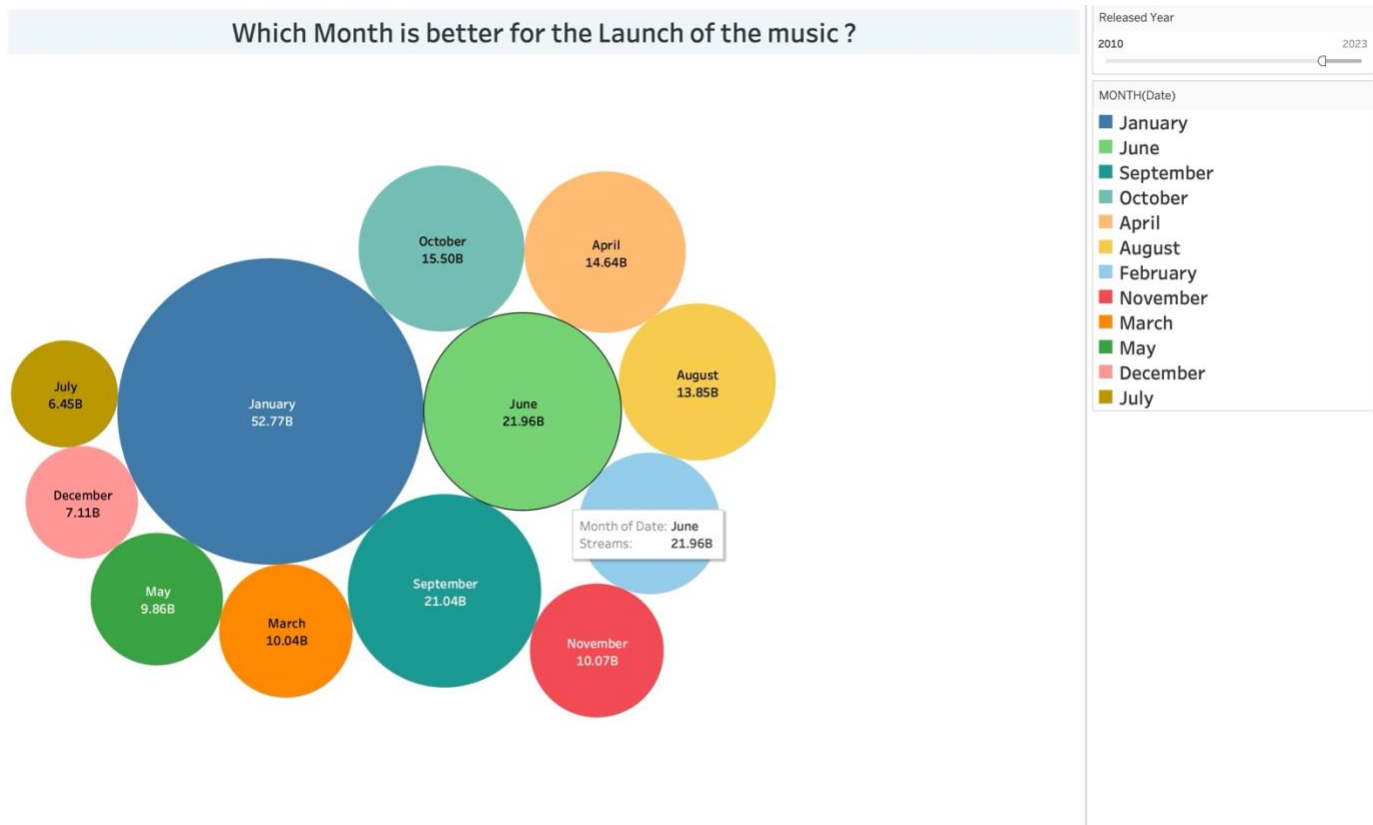
Prediction/Prescription:

1. These findings suggest that Sony Music should concentrate on investing in and marketing its top performers, since their proven success can result in sizable streaming figures and income.
2. Leading musicians working together could increase streams and draw in a variety of audience demographics.

3. Maximizing the impact of these great musicians on streaming platforms can be achieved by crafting marketing strategies that highlight their achievements and make use of their influence.
4. Post Malone and Swae Lee combination has best performed in getting the total sum of streams on Spotify. Justin Bieber and The Kid Laroï were at the second position to gather streams from 2010

VISUALISATION 2

Which Month is better for the Release of the Music?



(VIZ 2 from Tableau Workbook)

How I Did It:

- **Data Visualization Tool:** Created a bubble chart in Tableau.
- **Chart Elements:**
 - Each bubble represents a month.
 - Bubble size denotes the total number of streams.
- **Colour Coding:** Used Released Date filtered by month for colour.

- **Labelling:** Both SUM of Streams and Released Date were used for text labels.

In worksheet of Tableau, I have put SUM of Streams in the Size, and for the Colours I have put the Released Date which is further down filtered to the month. Then, both will be used in Text as well for labelling the graph. The Colour and the size distinction both helps in figuring out the best month for the launch of the music.

What It Tells Us:

1. **January** stands out with the largest bubble, indicating the highest streaming activity. This is a prime time to release new music and capitalize on peak user engagement.
2. **June and September** also show substantial streaming volumes, making them excellent secondary choices for music launches. If any project is ready to launch, then what are the second best options.
3. **July and December** have the smallest bubbles, suggesting lower streaming activity. These months may require special strategies to boost engagement

Prediction/Recommendation:

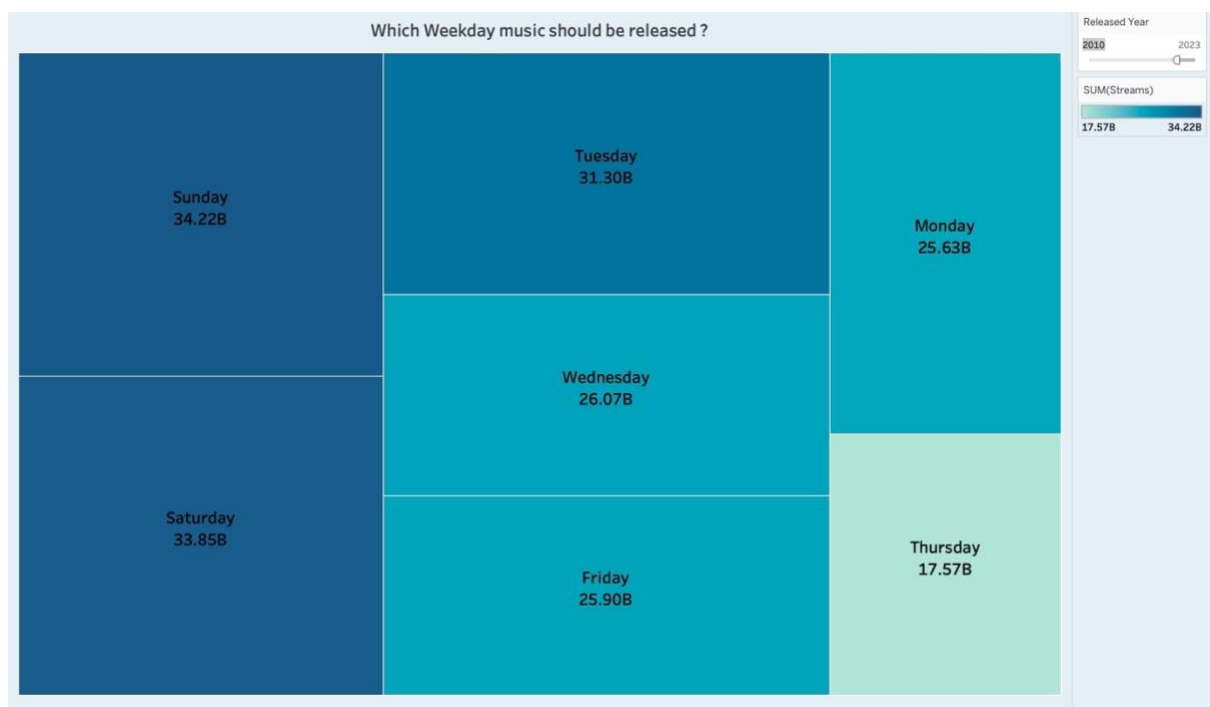
Based on these findings, here are some recommendations to optimize our music release schedule:

1. **Launch Major Releases in January:** This is when our audience is most active. Releasing new music during this peak period can significantly boost streams and visibility.

2. **Target June and September for Secondary Releases:** These months also show high engagement, making them ideal for follow-up releases or major artist projects.
3. **Special Campaigns for July and December:** Since these months have lower activity, we should consider exclusive releases, special promotions, or collaborations to attract more listeners. Think of it as an opportunity to stand out when there's less competition.

VISUALISATION 3

Which Weekday is Better for Launch of Music on Spotify?



How I Did It

- **Data Visualization Tool:** Created a tree map in Tableau to display streaming activity across different weekdays.
- **Chart Elements:**
 - Each rectangle represents a weekday.
 - Rectangle size and colour denote the total number of streams.
- **Colour Coding:** Used Released Year and SUM of Streams for colour gradient.
- **Labelling:** Weekday names and total streams labelled.

What It Tells Us

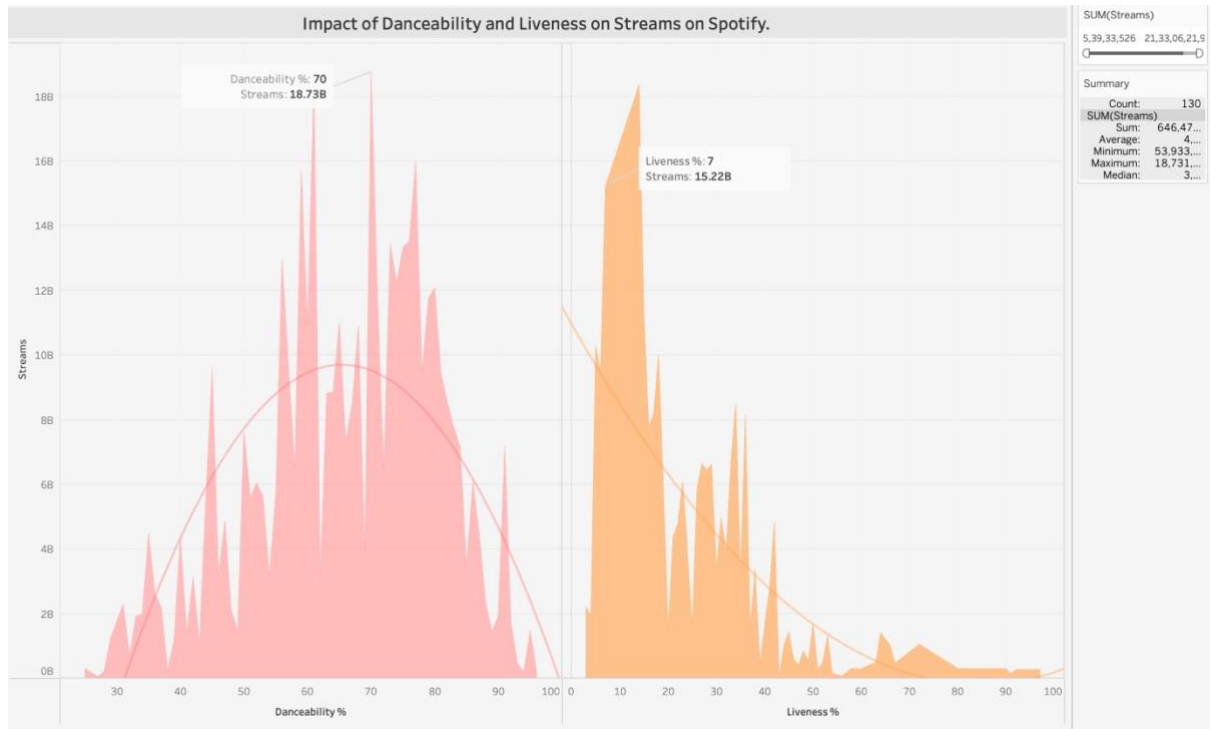
- Sunday: Largest rectangle, indicating the highest streaming activity; optimal for new music releases.
- Saturday: Substantial streaming volume, a good secondary choice for releases.
- Tuesday: Also shows significant streaming activity.
- Monday, Wednesday, and Friday: Moderate streaming activity.
- Thursday: Smallest rectangle, indicating the lowest streaming activity; may need special strategies to boost engagement.

Prediction/Prescription

- Primary Recommendation: Launch major releases on Sunday to maximize streams and visibility.
- Secondary Recommendation: Target Saturday and Tuesday for secondary or follow-up releases.
- Special Campaigns: Plan exclusive releases, promotions, or collaborations on Thursday to boost engagement.

VISUALISATION 4

Impact of Danceability and Liveness on the Streams on the streams of Spotify.



How I Did It:

Data Connection: Connected to the dataset (Cleaned_Spotify_Dataset.csv) in Tableau.

Data Types: Ensured danceability_%, liveness_%, and streams were set as numeric columns.

Chart Setup:

- Placed SUM(Streams) in the Rows shelf.
- Placed Danceability % and Liveness % in the Columns shelf.
- Changed chart type to Area under Danceability %.
- Filters: Added filters for Released Year and SUM(Streams) for interactivity.

Customization:

- Titled the chart "Impact of Danceability and Liveness on Streams on Spotify."
- Customized tooltips to display detailed information on danceability, liveness, and stream counts.

What It Tells Us:

1. Danceability:

Tracks with a danceability percentage between 50% and 80%, especially around 70%, tend to receive more streams.

Songs that are somewhat to extremely danceable are more popular on Spotify.

2. Liveness:

Tracks with liveness percentages between 20% and 30% exhibit notable streams. Songs with a reasonable number of live features are linked to greater appeal.

Songs with very high liveness percentages (nearly 100%) have fewer streams, indicating less listener preference.

Prediction/Prescription:

1. Danceability:

Future song producers should aim for a danceability percentage between 50% and 80%, focusing on about 70%, as this range is associated with higher stream counts.

2. Liveness:

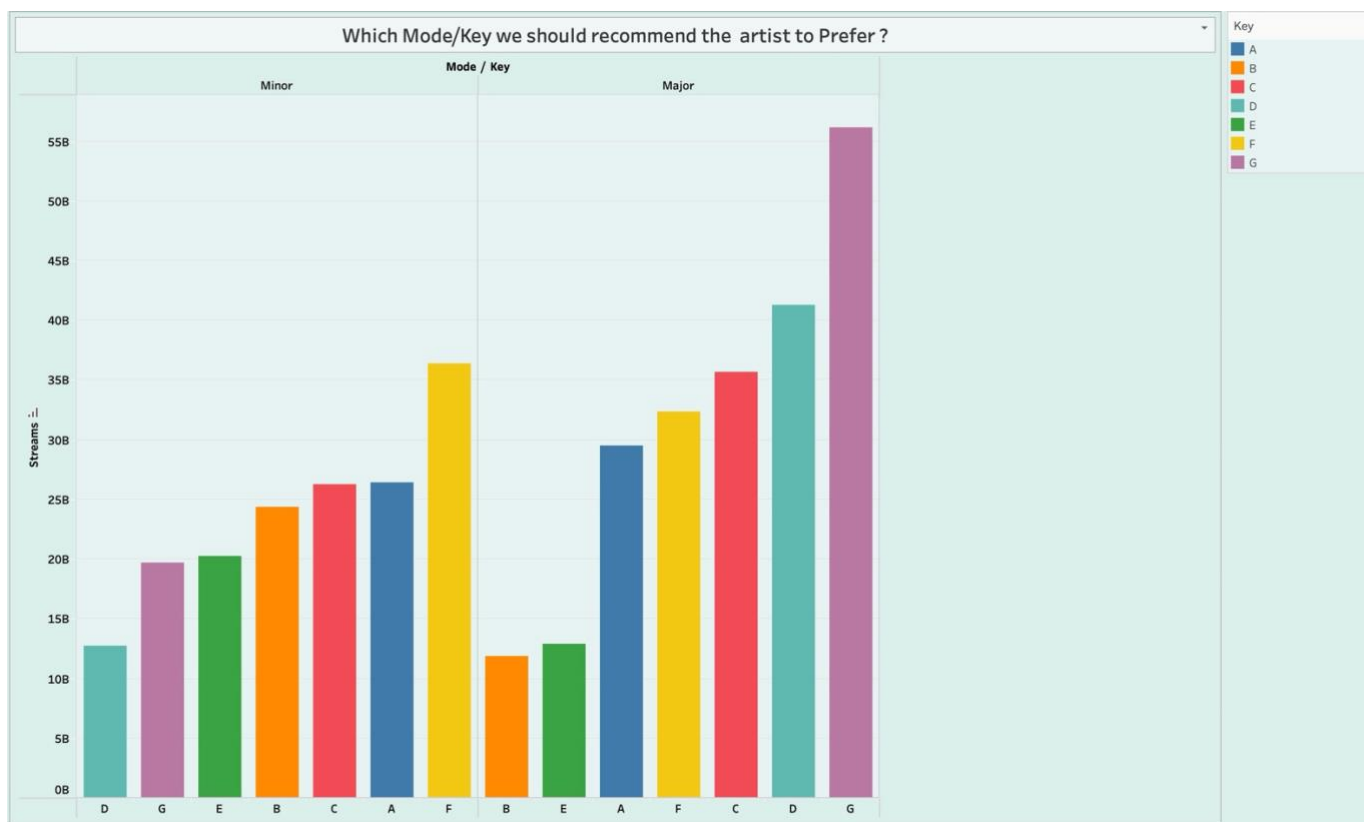
Adding mild live elements, aiming for a liveness percentage of 20% to 30%, can increase a track's appeal without reducing its broad appeal.

3. Marketing Strategy:

Include songs with these ideal audio qualities in high-visibility playlists.
Highlight the danceability of tracks within the identified optimal ranges in promotional campaigns to attract and retain listeners.

VISUALISATION 5

Which Mode/Key we should recommend the artist to prefer for more streams?



How I Did It

Tableau Connection:

- **Connection:** Connected to the dataset Cleaned_Spotify_Dataset.csv in Tableau to produce the visualization.
- **Data Configuration:** Verified that the data types for streams, mode, and key were correctly configured.
- **Creating the Composite Axis:** Dragged Mode to Columns and Key to Columns to create a composite axis that distinguishes between Major and Minor keys.
- **Displaying Total Streams:** Dragged SUM(Streams) to Rows to visualize the total number of streams for each key and mode combination.
- **Adding Interactivity:** Included a Released Year filter to allow users to focus on specific years.

What It Tells Us:

Visualization Insights: The visualization displays total streams for songs in various keys, separated by Major and Minor modes.

- **Major Keys:** Songs in Major keys generally have higher streams compared to those in Minor keys.
- Notably **Popular Major Keys:** G Major, C Major, and A Major, with G Major being the most popular.

- **Minor Keys:** A Minor has the highest stream count among Minor keys but is still lower than the top Major keys.

Interpretation:

- There is a clear preference for Major keys among listeners.
- Songs in Major keys are more likely to achieve higher popularity on Spotify.

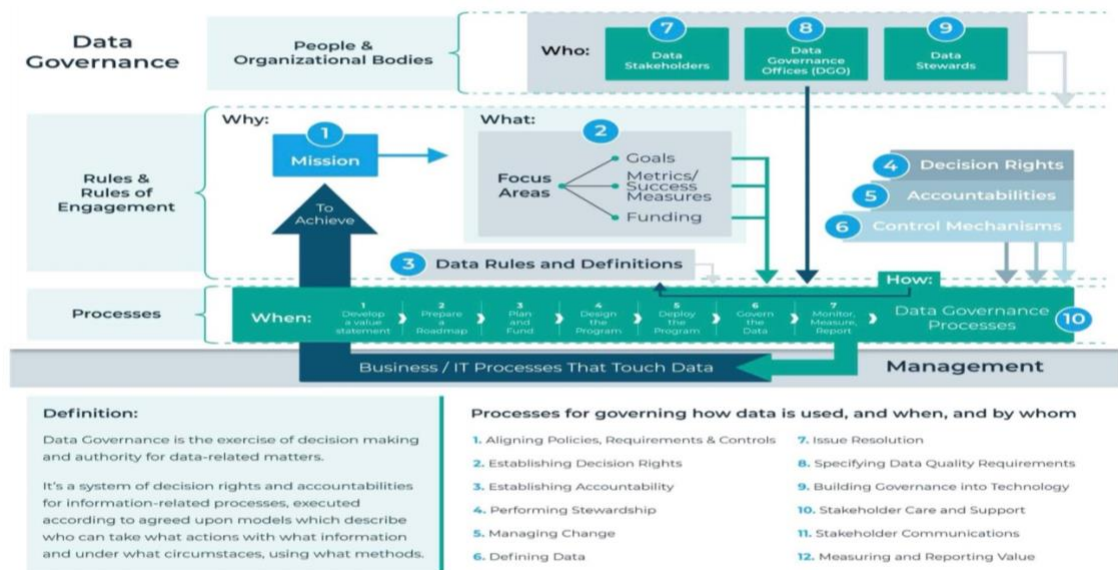
Prediction/Prescription:

Recommendations for Artists and Producers:

- **Focus on Major Keys:** To maximize potential for higher streams and popularity, artists should focus on composing songs in Major keys, especially G Major, C Major, and A Major.
- **Consider Minor Keys:** While Minor keys can still be used, emphasizing A Minor can be beneficial as it shows relatively higher popularity within Minor modes.
- **Future Productions:** Prioritize composing in the identified popular keys to enhance the chances of a song becoming a hit.

PART B -

Describe the four data analytics process for data governance



Four Data Analytics Processes for Data Governance

The four primary data analytics processes for data governance, based on the provided diagram, are:

1. Data Collection and Acquisition:

This entails obtaining information from diverse sources and making sure it complies with legal requirements and is accurate, pertinent, and up to date.

2. Data Storage and Management:

This involves ensuring data availability, preserving data quality over time, and securely storing data in databases or data warehouses. It involves developing metadata management, data recording, and data governance policies.

3. Data Analysis and Interpretation:

To obtain useful knowledge, this step involves examining the data. To convert unprocessed data into useful information, machine learning algorithms are applied, along with statistical analysis and data mining techniques.

4. Data Sharing and Visualization:

This involves presenting analysed data in a format that stakeholders can comprehend and utilize. Accuracy, clarity, and accessibility are among the data governance standards that are followed to when developing reports, dashboards, and data visualizations.

Features of Tableau Supporting Data Governance Principles

1. Accuracy:

Data Connectivity and Integration: Tableau facilitates substantial information collecting and integration by establishing connections with various data sources. *(Dhruvildave, 2021)*

2. Clarity:

Visualization Tools: To help present data in an easily understood manner, Tableau provides a variety of visualization options, including bar charts, scatter plots, heat maps, and more. Customizable dashboard being key.

3. Transparency:

Source Attribution: Using Tableau, users can easily credit the sources of their data, promoting transparency and enabling stakeholders to confirm the data's source.

ETHICAL VALUES

Ensuring the accuracy, reliability, and responsible communication of insights obtained from data is contingent upon the use of ethics in data visualization.

1. Fairness

Fairness is an ethical value that emphasizes treating everyone equally, without favouritism or discrimination.

Equal Access to Information: Make sure that the visualizations are accessible to all stakeholders, regardless of their background or technical expertise. Use clear and simple language and provide explanations for complex charts or terms. *(Quinn, 2024)*

Consistent Standards: Apply consistent standards in data processing and visualization across all datasets.

2. Accuracy (Aligned with Utilitarianism)

Unbiased Representation: Make sure that all groups have equal representation in data visualizations. For example, do not use color schemes or scales that suggest one group is better or worse than another when showing demographic data. *(ABC Religion & Ethics, 2020)*

Uniform Criteria: Use uniform guidelines for data processing and visualization for all datasets.

Ethical Value: In data visualization, accuracy refers to accurately displaying the data while avoiding distortion of any kind.

Application: Data Integrity: Make sure I'm using the most recent and correct data possible.

Correct Scaling: Make use of scales that fairly represent the size of the data. For example, to prevent inflating value discrepancies, make sure the y-axis of a bar chart begins at zero.

3. Transparency (Aligned with Common Good)

Application:

Unbiased Representation: Make sure that all groups have equal representation in data visualizations.

Uniform Criteria: Use uniform guidelines for data processing and visualization for all datasets.

Ethical Value: Transparency is being forthright about the sources of the data, the techniques used to obtain it, and any presumptions or restrictions on the data.

Techniques Notice of Disclosure: Describe the procedures utilized to gather and analyse the data.

Restrictions and Premises: Disclose any restrictions or presumptions in Transparency ensures that viewers are fully informed about the context and quality of the data, fostering trust and confidence in the visualization.

CONCLUSION

To summarise, a thorough analysis of Spotify's top-streamed songs in 2023 illustrates how data visualisation may revolutionise business decisions at Sony Music. We were able to uncover important information by using Tableau, like the popularity of musicians such as Ed Sheeran, Taylor Swift, and The Weeknd, the best times of year to release new music, and the importance of danceability and liveness for streaming performance. The significance of focusing marketing tactics to take advantage of peak streaming times and artist popularity is shown by these data.

We provide actionable advice for future music creation and marketing efforts by identifying ideal audio quality for maximizing streams, such as danceability percentages between 50% and 80% and liveness percentages between 20% and 30%. Furthermore, the examination of key modes indicates that writing in major keys—specifically, G Major, C Major, and A Major—should be prioritized to boost song popularity.

The correctness, dependability, and transparency of our results were guaranteed by ethical considerations and data governance rules, which promoted accountability and confidence. Sony Music can make improvements to its marketing strategy, predict future trends, and strengthen its competitive advantage in the ever-changing music industry by using these data-driven insights.

References

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